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ASSOCIATING ASSETS USING RFID-RF WIRELESS GATEWAYS

Abstract

A system is disclosed for tracking and monitoring sets of assets in or on a container using RFID-RF tags associated with the assets. The system comprises a specimen holder bin or similar container and a wireless tag affixed to the specimen holder bin. The specimen bin holder is configured to hold a plurality of assets. The wireless tag comprises a radio frequency identifier (RFID) circuit, a radio frequency (RF) circuit, a processor, a memory storing an identifier (ID) associated with the RFID circuit, and an energy source electrically connected to the RFID circuit, the RF circuit, and the processor. The wireless tag is configured to broadcast the identifier associated with the RFID circuit responsive to an interrogation signal by one or more gateways associated with the system. In some embodiments, each asset of the plurality of assets additionally comprises an RFID tag.

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Background/Summary

[0001] This application is a continuation of pending U.S. patent application Ser. No. 18/083,550, filed on Dec. 18, 2022, which is a divisional of U.S. patent application Ser. No. 16/953,238, filed on Nov. 19, 2020, now U.S. Pat. No. 11,907,796. U.S. patent application Ser. No. 16/953,238 claims priority to U.S. Provisional Patent Application No. 62/937,501, filed Nov. 19, 2019. All of the above-referenced applications are incorporated herein in their entirety.

TECHNICAL FIELD

[0002] Embodiments disclosed herein generally relate to wireless sensors and communications devices and systems for tracking and monitoring assets.

BACKGROUND

[0003] Wireless sensors and communications devices and systems have been used for tracking and monitoring sets of items in or on a container. For example, radio frequency identifier (RFID) tags conventionally are used in the medical and chemical fields to store and track containers of specimens, such as medical specimens and chemical specimens.

SUMMARY

[0004] RFID tags and barcode tags have limited or no active communications capabilities. For example, a RFID tag typically is limited to storing a unique identifier in memory and wirelessly transmitting that unique identifier periodically or on demand. Barcodes on the other hand are passive tags without any active functionality; instead, a barcode is read by an optical scanner with line-of-sight visibility of the barcode. Given their limitations, RFID and barcode technologies are unable to perform advanced communications and locationing functions.

[0005] In an aspect, a system is disclosed for tracking and monitoring sets of items in or on a container using RFID-RF tags associated with the items. In an embodiment, the system comprises a specimen holder bin comprising a plurality of slots and a wireless tag affixed to the specimen holder bin, e.g., by an adhesive or other means. The plurality of slots are sized to receive a respective specimen holder (e.g., vials). The wireless tag comprises a radio frequency identifier (RFID) circuit, a radio frequency (RF) communication circuit (e.g., a Bluetooth communication interface), a processor, a memory storing an identifier (ID) associated with the RFID circuit, and an energy source electrically connected to the RFID circuit, the RF circuit, and the processor. In some embodiments, the wireless tag encapsulates the RFID circuit, the RF circuit, the processor, the memory, and the energy source. The system may further comprise a plurality of specimen holders in each of the plurality of slots, wherein each specimen holder comprises a respective RFID tag affixed to an exterior surface of the specimen holder, e.g., by an adhesive or other means.

[0006] In other embodiments, the system comprises a plurality of items, each item comprising a wireless tag affixed to the respective item. Each wireless tag comprises an RFID circuit, a RF circuit, a processor, a memory storing an ID associated with the RFID circuit, and an energy source electrically connected to the RFID circuit, the RF circuit, and the processor. The wireless tags are affixed to the respective item using, for example, an adhesive or other means.

[0007] In some embodiments, the system further comprises a gateway operable to broadcast an interrogation signal to the wireless tag. The RFID tag in the wireless circuit is operable to, in response, broadcast the identifier associated with the RFID circuit to the gateway. In some embodiments, the gateway is further operable to broadcast an interrogation signal to the wireless tags, and RFID circuits in the respective RFID tags are operable to broadcast identifiers associated

with the specimen holders or the items in response, such that each specimen holder or each item of the plurality of specimen holders or items may be accounted for, tracked, and monitored.

[0008] In some embodiments, the gateway is further operable to broadcast an advertising packet to one or more RF circuits of the wireless tags. In response to receipt of the advertising packet, the one or more RF circuits are operable to establish a communication connection with the gateway. In some embodiments, the RF circuit is a Bluetooth Low Energy circuit, and the communication connection is a Bluetooth connection.

[0009] The following disclosure features methods, apparatus operable to implement the methods described herein, and computer-readable media storing computer-readable instructions causing a computer to implement the methods described herein.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIGS. 1A and 1B are perspective diagrams of two example specimen bins that includes a plurality of slots for holding respective specimen holders labeled with different wireless tags, according to some embodiments.

[0011] FIG. 2 is a cross-sectional diagram of an example RFID tag, according to some embodiments.

[0012] FIG. 3A is a schematic view of RFID/RF tag components of an example wireless transducer circuit, according to some embodiments.

[0013] FIG. 3B is a diagrammatic top view of multiple segments of an example wireless tag platform containing respective embedded wireless transducing circuits, according to some embodiments.

[0014] FIG. 4A is a diagrammatic cross-sectional side view of a wireless tag that includes a printed flexible battery, according to some embodiments.

[0015] FIG. 4B is diagrammatic cross-sectional side view of a wireless tag that includes a cell battery, according to some embodiments.

[0016] FIG. 5 is a flow diagram of a method of labeling specimen holders and a specimen holder bin with unique identifiers, according to some embodiments.

[0017] FIG. 6 is a diagrammatic view of correlated specimen holder and bin identifier entries in a database, according to some embodiments.

[0018] FIG. 7 is a diagrammatic view of bins of vials being loaded into a vehicle after passing through an RFID/RF gateway, according to some embodiments.

[0019] FIG. 8A is a schematic diagram of a package passing through an RFID/RF gateway operable to read and correlate asset identifiers, according to some embodiments.

[0020] FIG. 8B is a block diagram of example components of the multimode wireless scanner device shown in FIG. 8B and an example database storing associations between correlated identifiers, according to some embodiments.

[0021] FIG. 9 is an example flow diagram of a process of reading, correlating, and storing asset identifiers, according to some embodiments.

[0022] FIG. 10 is an example physical premises environment, according to some embodiments.

[0023] FIG. 11 is a diagrammatic view of an example of a network environment supporting communications with segments of an adhesive tape platform, according to some embodiments.

[0024] FIG. 12 is a diagrammatic view of a hierarchical communications network, according to some embodiments.

[0025] FIG. 13 is a flow diagram of a method of creating a hierarchical communications network, according to some embodiments.

[0026] FIGS. 14A-14E are diagrammatic views of exemplary use cases for a distributed agent

operating system, according to some embodiments.

[0027] FIG. **15A-15B** are diagrammatic top views of a length of an example tracking adhesive product, according to some embodiments.

[0028] FIG. **16** is a block diagram of an example computer apparatus, according to some embodiments.

[0029] In the following description, like reference numbers are used to identify like elements.

Furthermore, the drawings are intended to illustrate major features of exemplary embodiments in a diagrammatic manner. The drawings are not intended to depict every feature of actual embodiments nor relative dimensions of the depicted elements, and are not drawn to scale.

DETAILED DESCRIPTION

Introduction

[0030] The present invention is not limited in any way to the embodiments described herein.

Instead, the embodiments described below are merely examples of the invention. Therefore, the structural and functional details disclosed herein are not to be construed as limiting the claims. The disclosure merely provides bases for the claims and representative examples that enable one skilled in the art to make and use the claimed inventions. Furthermore, the terms and phrases used herein are intended to provide a comprehensible description of the invention without being limiting.

[0031] As used herein, the term “or” refers an inclusive “or” rather than an exclusive “or.” In addition, the articles “a” and “an” as used in the specification and claims mean “one or more” unless specified otherwise or clear from the context to refer the singular form.

[0032] The term “data object” refers to an addressable data file or a subset thereof. The term “metadata” include information about data objects or characteristics thereof. The terms “module,” “manager,” and “unit” refer to hardware, software, or firmware, or a combination thereof.

[0033] The terms “wireless tag,” “wireless node,” or “tape node” may be used interchangeably in certain contexts, and refer to an adhesive tape platform or a segment thereof that is equipped with sensor, processor, memory, energy source/harvesting mechanism, and wireless communications functionality, where the adhesive product has a variety of different form factors, including a multilayer roll or a sheet that includes a plurality of divisible adhesive segments. Once deployed, each tape node or wireless node can function, for example, as an adhesive tape, label, sticker, decal, a card, a wireless tag, a tracking device, or the like, and as a wireless communications device. A “peripheral” tape node or wireless node, also referred to as an outer node, leaf node, or terminal node, refers to a node that does not have any child nodes.

[0034] In certain contexts, the terms “wireless tracking system,” “hierarchical communications network,” “distributed agent operating system,” and the like are used interchangeably herein to refer to a system or network of wireless nodes.

Monitoring Sets of Assets Using RFID-RF Tags

[0035] Embodiments described herein relate to wireless sensors and communications tags, devices, and systems for tracking and monitoring assets. In particular, embodiments herein relate to methods and systems for tracking and monitoring assets being contained or transported within or as part of a container, e.g., specimens, vials, and the like contained within a sample bin. In conventional methods, containers having multiple assets are often tracked as a single item; however, this makes it difficult to identify whether individual assets are lost, damaged, or otherwise exposed to harmful conditions. It is valuable for each asset within a container to be accounted for during transportation, shipment, some other movement, storage, other phases of the asset, or some combination thereof in order to ensure that individual assets are not lost or damaged.

[0036] FIG. **1A** shows a diagrammatic perspective view of a specimen sample bin **10** that includes a plurality of slots, including slot **12**, for receiving specimen sample holders **14** (also referred to herein as “vials”). The vial **14** shown in FIG. **1A** includes a stopper **16** and an RFID tag **18**. The vial **14** is an example of an asset, and the specimen sample bin **10** is an example of a container for one or more assets including the vial **14**. In some embodiments, the specimen sample bin **10** holds

multiple vials **14**. In other embodiments, the specimen sample bin **10** may hold a single vial **14**. In the illustrated example, the vial **14** may be any type of suitable container, including glass or plastic vials. The vial **14** may contain, for example, medication, vaccinations, chemicals or other liquids, gases, solids, or any other suitable specimen. In other embodiments, assets may be different, and a container for the assets may be different, e.g., the assets may be petri dishes stored within a sample holder. The RFID tag **18** on the vial **14** may be any type of active or passive RFID tag. In the example embodiment shown, the RFID tag **18** has a rectangular form factor and is adhered to a vial **14** by an adhesive. In other examples, the RFID tag **18** has a shape corresponding to a shape other than a rectangle, such as a triangle, a circle, or some other shape. In other examples, the RFID tag **18** may have other, irregular, or specialized form factors, such as a luggage tag form factor wherein the RFID tag is able to be wound around a portion of an asset and adhered to itself or the asset. In other examples, the RFID tag **18** may be specialized to be adhered to a particular asset, e.g., to a narrow or irregular portion of an asset, or to be looped through or tied to a portion of the asset. In other examples, the form factor of the RFID tag **18** is that of an adhesive tape, having one or more adhesive surfaces or adhesive portions of surfaces. In other examples, the form factor may be smaller (e.g., so as to fit into narrow or tight areas of an asset) or larger (e.g., so as to be clearly visible on large assets), and may comprise additional or different mechanisms for adhering or affixing to assets.

[0037] Each of the RFID tag **18** and the RFID/RF tag **20** may be an active RFID tag or a passive RFID tag, according to some embodiments. An active RFID tag has an internal power source for transmitting a unique identifier stored in memory. A passive RFID tag may not include an internal power source and may respond to a radio frequency (RF) interrogation signal by transmitting the unique identifier stored in a memory device embedded in the tag. The specimen sample bin **10** also includes a RFID/RF tag **20** adhered to a side wall **19** of the specimen sample bin **10**. In the example embodiment of FIG. **1A**, the RFID/RF tag **20** is adhered to a side wall **19** of the specimen sample bin **10** by an adhesive. In other examples, the RFID/RF tag **20** may instead or additionally be affixed to the specimen sample bin **10** using, for example, a specialized form factor, a luggage tag, or another means, as described previously in conjunction with the RFID tag **18**. The RFID/RF tag **20** includes an RFID circuit and a radio frequency (RF) circuit. In some examples, the radio frequency (RF) circuit is a Bluetooth® Low Energy communications circuit.

[0038] FIG. **1B** shows a diagrammatic perspective view of a specimen sample bin **11** that is identical to the specimen sample bin **10** (shown in FIG. **1**), except the RFID/RF tag **20** that is adhered to the specimen sample bin **10** has been replaced by an RF tag **21** (e.g., a Bluetooth® LE communications tag wireless tag).

[0039] FIG. **2** shows a diagrammatic cross-sectional view of an example of the RFID tag **18** shown in FIGS. **1A** and **1B**. The RFID tag **18** includes a cover **22**, which may be formed of a RFID printer compatible laminate. The cover **22** may be laminated to a backing layer **23**. The backing layer **23** is disposed over the RFID circuit layer **24**, which includes a memory device **25** that stores a unique radio frequency identifier **26**. The circuit layer **24** is disposed over an adhesive layer **27**, which is positioned over a release layer **28**. The release layer **28** may be removably adhered to the bottom surface of the adhesive layer.

[0040] In some embodiments, the form factor of the RFID tag **18** and/or of the RFID/RF tag **20** is a flexible adhesive tape platform comprising a flexible polymer layer encapsulating one or more electronic components (as discussed further in FIGS. **3A-3B**) of the wireless tag, so as to enable the application of bending, torqueing, pressing, or other forces that may be applied to the flexible adhesive tape platform. For example, as in the luggage tag form factor, the flexible form factor may be bent along an axis to adhere to itself, e.g., around a wire, cord, lever, or handle of an asset. The flexible adhesive tape platform may further comprise one or more flexible film layers having reflective surfaces or reflective surface coatings and/or paper substrates. Example compositions for flexible film layers include polymer films, such as polyester, polyimide, polyethylene terephthalate

(PET), and other plastics. The flexible adhesive tape platform may further comprise one or more pressure-sensitive (e.g., silicon-based) adhesive layers or flexible epoxy layers. In other examples, other compositions and layers may be used, e.g., insulating layers or materials, colored layers or materials, light-reflective layers or materials, light-absorbing layers or materials, and/or light-emitting coatings, layers, or materials. In some embodiments, one or more layers of the form factor are adhered together, e.g., the form factor comprises a first adhesive layer between the cover **22** and the backing layer **23**, and further comprises a second adhesive layer between the backing layer **23** and the RFID circuit layer **24** of the RFID tag **18** and/or the RFID/RF tag **20**.

[0041] FIG. **3A** shows a block diagram of the components of an example RFID/RF tag circuit **32** that includes two or more wireless communications systems **34**, **36**. One or more of the RFID tag **18** and the RFID/RF tags **20** and **21** shown in FIGS. **1A**, **1B**, and **2** may include one or more embodiments of the RFID/RF tag circuit **32**. For example, the RFID tag circuit **32** may be disposed in the RFID/RF tags **20**, **21** shown in FIGS. **1A** and **1B**, according to some embodiments. Example communication systems **34**, **36** include a GPS system that includes a GPS receiver circuit **38** (e.g., a receiver integrated circuit) and a GPS antenna **41**, and one or more wireless communication systems each of which includes a respective transceiver circuit **47** (e.g., a transceiver integrated circuit) and a respective antenna **45**. Example wireless communication systems include a cellular communication system (e.g., GSM/GPRS), a Wi-Fi communication system, an RF communication system (e.g., LoRa), a Bluetooth communication system (e.g., a Bluetooth Low Energy system), a Z-wave communication system, and a ZigBee communication system. The wireless tag circuit **32** also includes a processor **37** (e.g., a microcontroller or microprocessor), one or more energy storage devices **38** (e.g., non-rechargeable or rechargeable printed flexible battery, conventional single or multiple cell battery, and/or a super capacitor or charge pump), one or more transducers **40** (e.g., sensors and/or actuators, and, optionally, one or more energy harvesting transducer components). Example communications systems include an RFID communications system **34** and an RF communications system **36**. Example RF communication systems include a Bluetooth® communication system (e.g., a Bluetooth® Low Energy system), a Z-wave communication system, a ZigBee communication system, and a LoRa communication system. The RFID/RD tag circuit **32** also includes a processor **37** (e.g., a microcontroller or microprocessor), one or more energy storage devices **38** (e.g., non-rechargeable or rechargeable printed flexible battery, conventional single or multiple cell battery, and/or a super capacitor or charge pump), a timer **39**, and one or more transducers **40** (e.g., sensors and/or actuators, and, optionally, one or more energy harvesting transducer components). In some examples, the conventional single or multiple cell battery may be a watch style disk or button cell battery that is electrically connected to an electrical terminal structure (e.g., a metal clip) that electrically connects the electrodes of the battery to contact pads on the wireless tag circuit **32**. Some embodiments also include a GPS system that includes a GPS receiver circuit (e.g., a receiver integrated circuit) that determines the current location of the RFID/RF tag circuit **32**.

[0042] Examples of sensing transducers **40** include a capacitive sensor, an altimeter, a gyroscope, an accelerometer, a temperature sensor, a strain sensor, a pressure sensor, a piezoelectric sensor, a weight sensor, an optical or light sensor (e.g., a photodiode or a camera), an acoustic or sound sensor (e.g., a microphone), a smoke detector, a radioactivity sensor, a chemical sensor (e.g., an explosives detector), a biosensor (e.g., a blood glucose biosensor, odor detectors, antibody based pathogen, food, and water contaminant and toxin detectors, DNA detectors, microbial detectors, pregnancy detectors, and ozone detectors), a magnetic sensor, an electromagnetic field sensor, and a humidity sensor. Examples of actuating (e.g., energy emitting) transducers **50** include light emitting components (e.g., light emitting diodes and displays), electro-acoustic transducers (e.g., audio speakers), electric motors, and thermal radiators (e.g., an electrical resistor or a thermoelectric cooler).

[0043] In some examples, the RFID/RF tag circuit **32** includes a memory **42** for storing data,

including, e.g., profile data, state data, event data, sensor data, localization data, security data, and one or more unique identifiers (IDs) **44** associated with the RFID/RF Tag circuit **32**, such as a product ID, a type ID, and a media access control (MAC) ID, and control code **46** that includes instructions executable by the processor **36** to perform one or more tasks autonomously. The one or more unique IDs may include a globally unique identifier associated with the RFID/RF tag circuit, according to some embodiments. In some examples, the memory **42** may be incorporated into one or more of the processors **36** or transducers **50** or may be a separate component that is integrated in the RFID/RF tag circuit **32**. The control code **46** typically is implemented as programmatic functions or program modules that control the operation of the RFID/RF tag circuit **32**, including a node communication manager that manages the manner and timing of tape node communications, a node power manager that manages power consumption, and a node connection manager that controls whether connections with other nodes are secure connections (e.g., connections secured by public key cryptography) or unsecure connections, and a node storage manager that securely manages the local data storage on the RFID/RF tag circuit **32**. In some examples, a node connection manager (not shown) ensures the level of security required by the end application and supports various encryption mechanisms. In some examples, a node power manager and communication manager (not shown) work together to optimize the battery consumption for data communication. In some examples, execution of the control code by the different types of nodes described herein may result in the performance of similar or different functions.

[0044] FIG. **3B** is a top view of a tape platform **50**. The tape platform **50** includes multiple segments, each segment including a respective set of the components of the RFID/RF tag circuit **32**. In some embodiments, multiple segments contain respective sets of components that are identical and configured in the same way. In some other embodiments, however, multiple segments contain respective sets of components that are different and/or configured in different ways. For example, different ones of the segments have different sets or configurations of tracking and/or transducing components in their respective RFID/RF tag circuit **32** that are designed and/or optimized for different applications. Alternatively, different sets of segments of the platform **50** may have different ornamentations (e.g., markings on the exterior surface of the platform) and/or different dimensions.

[0045] An example method of fabricating the tape platform **50** according to a roll-to-roll fabrication process is described in connection with FIGS. **6**, **7A**, and **7B** of U.S. patent application Ser. No. 15/842,861, filed Dec. 14, 2015, now U.S. Pat. No. 10,262,255, the entirety of which is incorporated herein by reference.

[0046] The instant specification describes an example system of wireless tags and adhesive tape platforms (also referred to herein as “tape nodes”) that can be used to implement a low-cost wireless network infrastructure for performing monitoring, tracking, and other asset management functions relating to, for example, parcels, persons, tools, equipment and other physical assets and objects. The example system includes a set of three different types of tape nodes that have different respective functionalities and different respective cover markings that visually distinguish the different tape node types from one another. In one non-limiting example, the covers of the different tape node types are marked with different colors (e.g., white, green, and black). In the illustrated examples, the different tape node types are distinguishable from one another by their respective wireless communications capabilities and their respective sensing capabilities.

[0047] FIG. **4A** shows a cross-sectional side view of a portion of a segment of a wireless tag **60** that includes tracking components **62**. The components **62** may include one or more embodiments of the RFID/RF tag circuit **32** shown in FIGS. **3A** and **3B**. The wireless tag segment **60** includes a flexible substrate **64** with an adhesive layer on its top surface and an optional adhesive layer on its bottom surface. If the bottom adhesive layer is present, a release liner (not shown) may be (removably) adhered to the bottom surface of the adhesive layer. The adhesive layer **66** bonds the flexible substrate **64** to a bottom surface of a flexible circuit **68** that includes one or more wiring

layers (not shown) that connect a processor, a circuit (e.g., a wireless receiver circuit, wireless transmitter circuit, or wireless transceiver circuit), an antenna, and other components (e.g., one or more sensors) in the device layer to each other and to the flexible battery **70** and, thereby, enable the tracking and other functionalities of the tracking adhesive product segment **60**. A flexible polymer layer **72** encapsulates the device layer and thereby reduces the risk of damage that may result from the intrusion of contaminants and/or liquids (e.g., water). The flexible polymer layer **72** also planarizes the device layer, which distributes forces generated in, on or across the tracking adhesive product segment **60** so as to reduce potentially damaging asymmetric stresses that might be caused by the application of bending, torqueing, pressing, or other forces on the tracking adhesive product segment **231**. A flexible cover **74** is bonded to the planarizing polymer **84** by an adhesive layer **82**.

[0048] In some embodiments, the flexible polymer layer **72** may comprise one or more flexible film layers having reflective surfaces, reflective surface coatings, and/or paper substrates. Example compositions for the flexible polymer layer **72** comprise, for example, polyester, polyimide, polyethylene terephthalate (PET), and other plastics. The flexible adhesive tape platform may further comprise one or more pressure-sensitive (e.g., silicon-based) adhesive layers or flexible epoxy layers. In other examples, other compositions and layers may be used, e.g., insulating layers or materials, colored layers or materials, light-reflective layers or materials, light-absorbing layers or materials, and/or light-emitting coatings, layers, or materials. In some embodiments, one or more layers or components of the wireless tag **60** may be rigid such that the adhesive tape platform is fully rigid or rigid along one or more axes. In some embodiments, the wireless tag **60** may comprise modules or portions that are mechanically or chemically affixed or adhered together and may be removeable, e.g., a module comprising the flexible battery **70** or other components.

[0049] FIG. **4B** shows a cross-sectional side view of an alternative example **78** of the segment of the wireless tag **78** shown in FIG. **4A**. The only difference between this alternative example **78** and the example **60** shown in FIG. **4A** is that the flexible battery **70** has been replaced by a conventional single or multiple cell battery **80** (e.g., a watch style disk or button cell battery) and associated electrical connection apparatus **83** (e.g., a metal clip) that electrically connects the electrodes of the battery **80** to contact pads on the flexible circuit **68**. In alternative examples, a segment of a wireless tag **78** may additionally or instead comprise removable batteries or batteries that may be recharged using a wireless or wired connection.

[0050] FIG. **5** is a flow diagram of a process of labeling specimen holders (e.g., vials) and a specimen holder bin **10**, **11** with globally unique identifiers as the specimen holders are being loaded into the slots of the bin **10**, **11**. The process includes labeling the bin with an RFID/RF tag by adhering or attaching an embodiment of a RFID/RF tag **20** to the bin **10**, **11** (FIG. **5**, block **500**). In some embodiments, the RFID/RF tag has an adhesive layer that attaches the RFID/RF tag to the bin. In other embodiments as discussed in conjunction with FIG. **1**, other methods of adhering or attaching the RFID/RF tag to the bin may be used, e.g., a luggage tag form factor. The bin RFID/RF tag is read by an RFID/RF scanner to obtain a first bin RF identifier and a second bin RF identifier from the encoded the RFID/RF tag (FIG. **5**, block **502**). The first bin identifier may be a RFID identifier. The second bin RF identifier is a different identifier from the first bin RF identifier. In some embodiments, the second bin RF identifier (RF ID) is a unique identifier of a communication interface of the RFID/RF tag. For example, the second bin RF identifier may be a unique identifier of a Bluetooth communication interface of the RFID/RF tag. In some embodiments, the first bin RF identifier and the second bin RF identifier are each globally unique. The RF scanner associates the first bin RF identifier with the second bin RF identifier and stores the correlated identifiers and their association in a database (FIG. **5**, block **504**).

[0051] The process continues by labeling a new specimen holder with a RFID tag by attaching or adhering the RFID tag to the specimen holder (FIG. **5**, block **506**). In some embodiments, the RFID tag on the specimen holder has a globally unique RF identifier. The RFID tag on the specimen

holder is identified by the RF scanner by comparing a current set of scanned RFID tags including the new specimen holder with a preceding set of RFID tags that have been read and stored in the database (FIG. 5, block **508**). Assuming all of the RFID tags previously deployed have been scanned at some point in time, the newly deployed RFID tag will be the RFID tag that does not match any of the RFID tags in the prior set of scanned RFID tags. Next, the RFID/RF scanner associates the RF identifier of the RFID tag on the new specimen holder with the identifiers of the RFID/RF tag affixed to the bin and stores the associations in the database (FIG. 5, block **510**). If there is an opening in the specimen holder bin for receiving a new specimen holder, the process continues by labeling a new specimen holder with a globally unique RFID tag and inserting the new specimen holder into an open slot in the bin (FIG. 5, block **512**). The system executing the process may determine that there is an open slot by counting the number of specimen holders (based on the RF identifiers of their respective RFID tags) associated with the bin and comparing the number with a number of total slots in the bin. If the current bin is filled, the process of storing specimen holders in the current bin is terminated (FIG. 5, block **514**).

[0052] In other embodiments, the RFID/RF scanner only reads the first bin RF identifier. In this case, the system associates each specimen holder RF identifier of specimen holders loaded into the bin with the first bin RF identifier.

[0053] FIG. 6 shows a diagrammatic view of a database **610** that contains globally unique correlated specimen holder (vial) and bin identifier entries in a database. In an embodiment, the identifier entries are read and entered to the database using the method discussed in conjunction with FIG. 5, e.g., by an RF scanner. In the illustrated embodiment, the database **610** shows an example set of vial RF IDs (e.g., RFID A001, RFID A002, . . . RFID A015) correlated with a globally unique first bin RFID identifier (e.g., RFID B001) and a globally unique second bin RF identifier (e.g., BLEID C001) that is associated with Bin 1. In other embodiments, additional or other information may be stored in the database **610**, and the information may be formatted differently. For example, the database **610** may additionally identify one or more of: a starting destination associated with a specimen holder and/or a bin; an ending destination associated with a specimen holder and/or a bin; one or more ambient conditions required by a specimen holder and/or bin; and the like. In another example, the database **610** may additionally identify related or linked specimen holder bins, e.g., if multiple bins contain similar assets, require similar conditions, or are being transported to a same destination. According to some embodiments, corresponding information including associated bin RF identifiers and vial RF identifiers for additional bins (e.g., specimen holder bin 2) and specimen holders may be stored in the database **610** for tracking each bin and their associated specimen holders.

[0054] FIG. 7 shows a warehouse containing items to be shipped. In the illustrated embodiment, the items are packages each containing one or more logistic items (e.g., bins of specimen vials). Each bin is associated with a globally unique first bin RF identifier and a globally unique second bin RF identifier, as discussed previously in conjunction with FIG. 6. Each asset within the respective bin is associated with an RFID/RF tag or RF tag. The RFID/RF tag or RF tag of each asset is associated with a corresponding first bin RF identifier or second bin RF identifier for the corresponding bin. In some embodiments, the items may further comprise, for example, a wireless tag comprising one or more sensors capturing data describing an environment of the items. In another example, the items may further comprise wireless tags configured to communicate to servers or networks **720** of the wireless tracking system or other wireless tags in the environment via other communications interfaces.

[0055] The warehouse environment comprises one or more RFID/RF gateways **712**, **714**. In embodiments, the one or more RFID/RF gateways are deployed throughout the warehouse environment such that assets cannot leave an area of the warehouse environment (e.g., a storage area, the warehouse building, etc.) without encountering an RFID/RF gateway. In other embodiments, the one or more RFID/RF gateways may be deployed differently, e.g., based on a

most likely trajectory through the warehouse environment. When the items pass through a set of RFID/RF gateways **712, 714**, each item is scanned by one or both of the RFID/RF gateways **712, 714** to ascertain the wireless tag identifiers of all the bins and vials that are loaded into a vehicle **718**. In an embodiment, the gateways **712, 714** transmit the identifiers of the scanned bins and vials over a network **720** to a server **722**, which is configured to access a database **716** and to detect if there are any missing bins or any vials that are missing from a bin or if there are any additional bins or vials that are not listed in the database **716**. In this process, the server application **724** compares the bin and vial IDs that were entered into the database **714** as the vials were initially loaded into the bins with the bin and vial IDs that are scanned by the RFID/RF gateways **712, 714**. If the server application **724** does not detect all of the vials or bins that were listed in the database, or if the server application **724** detect the identifiers of any vials or bins that are not listed in the database **716**, the server application **724** is configured to raise an alarm.

[0056] In an embodiment, the server application **724** transmits the alarm or notification to a client device of a user of the application or of the wireless tracking system. The alarm or notification may be, for example, a text message or phone call notification transmitted to a mobile phone or computing device. The alarm or notification may comprise information identifying an event (e.g., whether a vial or bin was not detected or whether an extraneous or unlisted vial or bin was detected) and may additionally comprise information describing a vial or bin associated with the event such that the erroneous vial or bin may be identified, located, and addressed quickly by a user of the wireless tracking system. The alarm or notification may prompt a human operator to inspect one or more of the items associated with the RF identifier that triggered the alarm.

[0057] In general, the items being tracked are not limited to bins and specimen vials. Instead, any type of asset, item, object, package, container, vehicle, animal, and any other thing that can be tagged can be tracked and located by the systems and methods described herein.

[0058] FIG. **8A** shows an example system **800** for reading and correlating identifiers relating to a package **802**. In the illustrated embodiment, the package **802** contains multiple logistic containers **804, 806** (e.g., boxes) that are marked with respective RFID tags **808, 810**, where each of the containers **804, 806** holds one or more assets. Although only two example containers **804, 806** are shown in FIG. **8A**, the package **802** may contain a different number of containers (e.g., more than two), according to some embodiments. In the illustrated embodiment, the package **802** is sealed with an adhesive tape product **811** that includes an embedded wireless Bluetooth® Low Energy (BLE) communications component **813** and an embedded RFID component **815**. As the package **802** passes between a set of RFID/RF gateways **812, 814**, each adhesive tape product **811** of the package **802** and RFID tag **808, 810** of the boxes **804, 806** is scanned by one or both of the RFID/RF gateways **812, 814** to ascertain the wireless tag identifiers of all the packages **802** and boxes **804, 806** that pass between the gateways.

[0059] In other embodiments, the boxes **804, 806** are carried on a pallet or other container (not shown), instead of being contained in the package **802**. In these embodiments, a RFID/RF tag **20** (FIG. **1A**) or a RF tag **21** (FIG. **1B**) is affixed to the pallet or other container and RFID tags **18** are affixed to the packages **804, 806**. In some embodiments, a single RFID/RF tag or a RF tag is used to identify the logistic container in or on which are grouped logistic entities that are individually labeled with RFID tags.

[0060] In the illustrated embodiment, the RF scanner components of the gateways **812, 814** are implemented by a Bluetooth® Low Energy (BLE) communications devices. The gateways **812, 814** are operable to read and correlate the scanned RFID and BLE identifiers stored in the memories of the RFID/BLE tags. The gateways **812, 814** transmit the identifiers of the scanned packages and boxes over a network **720** to a server **722**, which is operable to detect if there are any packages or boxes that are missing and detect if there are any additional packages or boxes that are not listed in the database **716**. In this process, the server application **724** compares the identifiers (IDs) of the packages and boxes that were entered into the database **714** as the boxes were initially

loaded into the packages with the package and box IDs that are scanned by the RFID/RF gateways **812, 814**. If the server application **724** does not detect all of the vials or bins that were listed in the database, or if the server application **724** detects any unknown identifiers of any boxes or packages that are not listed in the database **716**, the server application **724** is operable to raise an alarm. [0061] In some embodiments, one or both of the RFID/BLE gateways **812, 814** are implemented by a multimode wireless gateway **820** of the type shown in FIG. **8B**. The gateway **820** includes a first reader module **823** (e.g., RFID scanner) and a second reader module **825** (e.g., RF scanner, e.g., BLE scanner). The multimode wireless gateway **820** also includes a power supply **826** (e.g., a battery), a processor **828**, and an association module **830**.

[0062] The first reader module **823** may be, for example, a RFID reader module that is configured to interrogate the RFID tag **815** of the adhesive tape product **11** on the package **802** and the RFID tags **808, 810** on the boxes **804, 806** within the package **802**. The RFID tags **815, 808, 810** may be configured with a fixed packet of read-only data that can be transmitted to a RFID reader (e.g., RFID reader module **823**) within range of the RFID tags **815, 808, 810**. The RFID tags **815, 808, 810** typically can be reprogrammed with different data, as needed. When the RFID reader module **823** detects a RFID tag **822** that is within range and operates within the same frequency range of the RFID reader module **823**, the RFID reader module **2** may read the RFID tag **22**. In operation, data is transmitted on modulated radio frequency electromagnetic waves between the RFID reader module **823** and the RFID tags **808, 810, 815**. In this process, the RFID reader module **823** transmits an electric or magnetic field that is sensed by the RFID tags **215, 208, 210**. In response, the RFID tags **815, 808, 810** transmit data (including a globally unique identification number) that typically is stored in a microchip memory component associated with the RFID tags **815, 808, 810**. [0063] In general, the RFID tags **815, 808, 810** may be an active RFID tag or a passive RFID tag. Active RFID tags include local power sources (e.g., batteries) for sending data packets to a RFID reader module. Passive RFID tags, on the other hand, do not require any local power sources to transmit data packets to a RFID reader module; instead, passive RFID tags are powered by inductive or capacitive coupling between the RFID reader module and the RFID tag. In an example, a passive RFID tag is configured to couple to the magnetic fields generated by a RFID reader module. In this regard, each of the RFID reader module and the RFID tag includes a respective set of one or more electrically conducting coils. The RFID reader module uses its power source to generate an electric current in the set of coils to generate magnetic fields that induce a current in the set of coils in the RFID tag. The induced current powers the RFID tag to generate a wireless RFID signal that is transmitted to the RFID reader module. In another example, a passive RFID tag is configured to capacitively couple with a corresponding RFID reader module through capacitive coupling plates. In this process, the RFID reader module generates an alternating electric field that causes the RFID tag to transfer data to the RFID reader module. However, capacitively coupled RFID reader modules and tags can only transfer information across short distances and therefore are typically limited to near-field applications.

[0064] The second reader module **825** may be, for example, a wireless RF scanner device that is configured to communicate with the wireless RF tag **813** of the wireless flexible product **811**. Example wireless RF scanner devices include a Bluetooth scanner (e.g., a Bluetooth Low Energy scanner), a near field communication (NFC) scanner, a LoRaWAN scanner, and a cellular scanner. In an illustrative embodiment, a Bluetooth Low Energy (BLE) scanner is configured to locate and communicate with BLE adhesive product tags within the scanner's range. In this process, the BLE scanner **824** advertises its presence with a specific authentication identifier and credentials. When a BLE adhesive product tag receives data from the BLE scanner, the BLE adhesive product tag establishes a handshake with the BLE scanner on the corresponding advertisement channel. Then the BLE adhesive product tag hands off communication with the BLE scanner **824** to a data channel (e.g., a BLE data channel). The BLE adhesive product tag learns the BLE scanner's product identification number (PIN) and type identification number (TIN) and transmits that

information to a network service to let the network service know that the BLE scanner **824** is communicating with the BLE adhesive product tag. Scanners and peripheral adhesive product tags for LoRaWAN, cellular, ZigBee, and other wireless communications operate in accordance with respective communications protocols.

[0065] In some examples, the first reader module **823** and the second reader module **825** are integrated into a single multimode reader device **820** that is operable to perform functions of the RFID/BLE gateways **812**, **814**. For example, the first reader module **823** and the second reader module **825** may be integrated into an RFID/RF scanning of the type shown in blocks **812**, **814** in FIG. **8A** to communicate with and read data from the wireless flexible product **811**, including the RFID tags **815**, **818**, **810** on and in the package **802** and the RF tag **813** on the package **802** (i.e., the Bluetooth Low Energy).

[0066] FIG. **9** shows a flow diagram of an example process of associating identifiers for assets having different RFID tags. In an embodiment, in accordance with this process, a radio frequency (RF) reader is directed to read a first tag attached to an asset and including a first identifier (FIG. **9**, block **950**). The radio frequency reader is directed to advertise its presence to the first tag and establish a wireless communications channel with the first tag to retrieve the first identifier from the first tag. For example, the RF reader transmits an interrogation signal to the first tag. The first tag is configured to, responsive to receiving the interrogation signal, establish a wireless communication channel and transmit a stored first identifier to the RF reader.

[0067] A second reader is instructed to read a second tag attached to the asset and including a second identifier (FIG. **9**, block **952**). In an embodiment, the second RF reader emits a first set of electromagnetic waves directed toward the second tag. The first set of electromagnetic waves result in a second set of reflected electromagnetic waves comprising the second identifier associated with the second tag. In other embodiments, additional or different sources or methods for reading RFID or RF tags may be used.

[0068] In some embodiments, the first tag and the second tag are integrated into a single physical entity. For example the first the tag and the second tag may each be integrated into an adhesive tape product, like the adhesive tape product **813** shown in FIG. **8A**. In other embodiments, the first tag is physically separate from the second tag. In further embodiments, the first tag is attached to a first asset, and the second tag is attached to a second asset associated with the first asset.

[0069] An association between the first identifier and the second identifier is stored predicated on the first tag and the second tag satisfying a temporal or spatial proximity condition (FIG. **9**, block **954**). For example, the embodiment shown in FIG. **1B** includes an RF tag **21** on the bin **11** but does not include a RFID tag on the bin **11**. In this embodiment, the bin is identified by the temporal proximity of the read times of the RFID tags on the vials in the bin to the time when the RF tag **21** is read. If the read times are within an empirically determined threshold time difference (e.g., 1-2 seconds), the RFID tags **18** on the vials that are read by the RFID reader are associated with the RF tag on the bin **11** that is scanned by the RF scanner component of the gateways **812**, **814**.

[0070] In some examples, the association module **830** component of the multimode wireless gateway **820** determines and transmits respective chains of associations between the different identifiers in one or more tables of the database **816** shown in FIG. **8A**. The tables may be used by an asset management application **824** to track assets, monitor the status or state of a particular asset, and report the status and condition of an asset. In some embodiments, the association database **816** includes a table of identifiers organized in a set of rows. Each row of identifiers is associated with a respective asset. For example, row 1 may correspond to the identifiers that are associated with asset 1, and row 2 corresponds to the identifiers that are associated with asset 2, and so on. Each row of associated identifiers enables the supplier, for example, to generate a report of the location, status, and condition of the associated asset as the asset travels through a logistics network by associating, for example, a transmitted RFID data packet with the identifier of the corresponding wireless flexible product.

[0071] Referring to FIG. 10, in some embodiments, an RF wireless tag (e.g., the BLE tag **811**) executes coded instructions that includes a mode of operation in which the RF wireless tag **813** transmits advertising packets into the physical premises environment **1060** and receives reply signals from infrastructure wireless tags **1062**, **1064**, **1066** with unique identifiers (i.e., ID1, ID2, and ID3) and respective signal strengths (e.g., RSSI). For example, in some examples, the RF wireless tag **813** on the package **802** can transmit the advertising packets in the physical premises environment **1060** and receive reply packets from wireless tags **1062**, **1064**, **1066** in the environment **1060**. The RF wireless tag **813** includes coded instructions to triangulate its location relative to the locations of the wireless tags **1062**, **1064**, **1066** from measurements of signal strength of the beacon signals being transmitted by the infrastructure wireless tags **1062**, **1064**, **1066** from different locations in the environment. In some embodiments, the received signal strength indicator (RSSI) is used as a measure of the power present in a received radio signal, where the received RSSI power level corresponds to respective distances between the BLE tag **813** and the wireless tags **1062**, **1064**, **1066** in the physical premises environment **1060**.

[0072] In other embodiments, other measures of wireless signal strength may be used, including, for example, time-of-flight and time-of-arrival measures. In other embodiments, the RF wireless tag **813** may further receive or capture other data associated with the physical premises environment **1060**, e.g., environment data via one or more sensors. Further, in other embodiments, other entities in the physical premises environment **1060** may perform one or more of the computing processes to triangulate or otherwise determine a location associated with the RF wireless tag **813** (and the corresponding package **802**). For example, in an embodiment, the RF wireless tag **813** is configured to transmit measurements of signal strength of the beacon signals to an infrastructure wireless tag **1062**, **1064**, **1066**. The infrastructure wireless tag **1062**, **1064**, **1066** may perform a triangulation to determine the location of the RF wireless tag **813**, or may further transmit the received measurements to a server or cloud of the wireless tracking system to perform the computation. In some embodiments, the entity responsible for performing the location determination is selected based on, for example, available computational resources, energy levels associated with a battery of the entities, or the like.

Deployment of Tape Nodes

[0073] FIG. 11 shows an example network communications environment **1100** that includes a network **1102** that supports communications between one or more servers **1104** executing one or more applications of a network service **1108**, mobile gateways **1110**, **1112**, a stationary gateway **1114**, and various types of tape nodes that are associated with various assets (e.g., parcels, equipment, tools, persons, and other things). The example network communication environment **1100** may also be referred to as a wireless tracking system **1100**. In an embodiment, the tape nodes comprise the RFID tags and RF wireless tags as described in conjunction with FIGS. 1-10, or may comprise other types of tape nodes for communicating with the RFID tags and RF wireless tags. The nodes of the wireless tracking system **1100** may refer to the tape nodes, other wireless devices, the gateway devices, client devices, servers, and other components of the wireless tracking system **1100**. In some examples, the network **1102** includes one or more network communication systems and technologies, including any one or more of wide area networks, local area networks, public networks (e.g., the internet), private networks (e.g., intranets and extranets), wired networks, and wireless networks. For example, the network **1102** includes communications infrastructure equipment, such as a geolocation satellite system **1116** (e.g., GPS, GLONASS, and NAVSTAR), cellular communication systems (e.g., GSM/GPRS), Wi-Fi communication systems, RF communication systems (e.g., LoRa), Bluetooth communication systems (e.g., a Bluetooth Low Energy system), Z-wave communication systems, and ZigBee communication systems.

[0074] In some examples, the one or more network service applications **1106** leverage the above-mentioned communications technologies to create a hierarchical wireless network of tape nodes that improves asset management operations by reducing costs and improving efficiency in a wide

range of processes, from asset packaging, asset transporting, asset tracking, asset condition monitoring, asset inventorying, and asset security verification. Communication across the network is secured by a variety of different security mechanisms. In the case of existing infrastructure, a communication link the communication uses the infrastructure security mechanisms. In case of communications among tapes nodes, the communication is secured through a custom security mechanism. In certain cases, tape nodes can also be configured to support block chain to protect the transmitted and stored data.

[0075] A set of wireless tags (e.g., tape nodes) can be configured by the network service **1108** to create hierarchical communications network. The hierarchy can be defined in terms of one or more factors, including functionality (e.g., wireless transmission range or power), role (e.g., master tape node vs. peripheral tape node), or cost (e.g., a tape node equipped with a cellular transceiver vs. a peripheral tape node equipped with a Bluetooth LE transceiver). Tape nodes can be assigned to different levels of a hierarchical network according to one or more of the above-mentioned factors. For example, the hierarchy can be defined in terms of communication range or power, where tape nodes with higher power or longer communication range transceivers are arranged at a higher level of the hierarchy than tape nodes with lower power or lower range transceivers. In another example, the hierarchy is defined in terms of role, where, e.g., a master tape node is programmed to bridge communications between a designated group of peripheral tape nodes and a gateway node or server node. The problem of finding an optimal hierarchical structure can be formulated as an optimization problem with battery capacity of nodes, power consumption in various modes of operation, desired latency, external environment, etc. and can be solved using modern optimization methods e.g. neural networks, artificial intelligence, and other machine learning computing systems that take expected and historical data to create an optimal solution and can create algorithms for modifying the system's behavior adaptively in the field.

[0076] In some examples, the one or more network service servers **1104** communicate over the network **1102** with one or more gateways that are configured to send, transmit, forward, or relay messages to the network **1102** and activated tape nodes that are associated with respective assets and within communication range. Example gateways include mobile gateways **1110**, **1112** and a stationary gateway **1114**. In some examples, the mobile gateways **1110**, **1112**, and the stationary gateway **1114** are able to communicate with the network **1102** and with designated sets or groups of tape nodes.

[0077] In some examples, the mobile gateway **1112** is a vehicle (e.g., a delivery truck or other mobile hub) that includes a wireless communications unit **1116** that is configured by the network service **1108** to communicate with a designated set of tape nodes, including a peripheral tape node **1118** in the form of a label that is adhered to an asset **1120** contained within a parcel **1121** (e.g., an envelope), and is further configured to communicate with the network service **1108** over the network **1102**. In some examples, the peripheral tape node **1118** includes a lower power wireless communications interface and the wireless communications unit **1116** is implemented by a tape node that includes a lower power communications interface for communicating with tape nodes within range of the mobile gateway **1112** and a higher power communications interface for communicating with the network **1102**. In this way, the tape nodes **1118** and **1116** create a hierarchical wireless network of nodes for transmitting, forwarding, bridging, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape node **1118** and the network service **1108** in a power-efficient and cost-effective way.

[0078] In some examples, the mobile gateway **1110** is a mobile phone that is operated by a human operator and executes a client application **1122** that is configured by the network service **1108** to communicate with a designated set of tape nodes, including a master tape node **1124** that is adhered to a parcel **1126** (e.g., a box), and is further configured to communicate with the network service **1108** over the network **1102**. In the illustrated example, the parcel **1126** contains a first parcel labeled or sealed by a tape node **1128** and containing a first asset **1130**, and a second parcel labeled

or sealed by a tape node **1132** and containing a second asset **1134**. As explained in detail below, the master tape node **1124** communicates with each of the peripheral tape nodes **1128**, **1132** and communicates with the mobile gateway **1108** in accordance with a hierarchical wireless network of tape nodes. In some examples, each of the peripheral tape nodes **1128**, **1132** includes a lower power wireless communications interface and the master tape node **1124** is implemented by a tape node that includes a lower power communications interface for communicating with the peripheral tape nodes **1128**, **1132** contained within the parcel **1126**, and a higher power communications interface for communicating with the mobile gateway **1110**. The master tape node **1124** is operable to relay wireless communications between the tape nodes **1128**, **1132** contained within the parcel **1126** and the mobile gateway **1110**, and the mobile gateway **1110** is operable to relay wireless communications between the master tape node **1124** and the network service **1108** over the wireless network **1102**. In this way, the master tape node **1124** and the peripheral tape nodes **1128** and **1132** create a hierarchical wireless network of nodes for transmitting, forwarding, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape nodes **1128**, **1132** and the network service **1108** in a power-efficient and cost-effective way.

[0079] In some examples, the stationary gateway **1114** is implemented by a server executing a server application that is configured by the network service **1108** to communicate with a designated set **1140** of tape nodes **1142**, **1144**, **1146**, **1148** that are adhered to respective parcels containing respective assets **1150**, **1152**, **1154**, **1156** on a pallet **1158**. In other examples, the stationary gateway **1114** is implemented by a tape node that is adhered to, for example, a wall, column or other infrastructure component of the environment **1100**, and includes a lower power communications interface for communicating with tape nodes within range of the stationary gateway **1114** and a higher power communications interface for communicating with the network **1102**. In one embodiment, each of the tape nodes **1142-1148** is a peripheral tape node and is configured by the network service **1108** to communicate individually with the stationary gateway **1114**, which relays communications from the tape nodes **1142-1148** to the network service **1108** through the stationary gateway **1114** and over the communications network **1102**. In another embodiment, one of the tape nodes **1142-1148** at a time is configured as a master tape node that transmits, forwards, relays, or otherwise communicate wireless messages to, between, or on behalf of the other tape nodes on the pallet **1158**. In this embodiment, the master tape node may be determined by the tape nodes **1142-1148** or designated by the network service **1108**. In some examples, the tape node with the longest range or highest remaining power level is determined to be the master tape node. In some examples, when the power level of the current master tape node drops below a certain level (e.g., a fixed power threshold level or a threshold level relative to the power levels of one or more of the other tape nodes), another one of the tape nodes assumes the role of the master tape node. In some examples, a master tape node **1159** is adhered to the pallet **1158** and is configured to perform the role of a master node for the tape nodes **1142-1148**. In these ways, the tape nodes **1142-1148**, **1158** are configurable to create different hierarchical wireless networks of nodes for transmitting, forwarding, relaying, bridging, or otherwise communicating wireless messages with the network service **1108** through the stationary gateway **1114** and over the network **1102** in a power-efficient and cost-effective way.

[0080] In the illustrated example, the stationary gateway **1114** also is configured by the network service **1108** to communicate with a designated set of tape nodes, including a master tape node **1160** that is adhered to the inside of a door **1162** of a shipping container **1164**, and is further configured to communicate with the network service **1108** over the network **1102**. In the illustrated example, the shipping container **1164** contains a number of parcels labeled or sealed by respective peripheral tape nodes **1166** and containing respective assets. The master tape node **1116** communicates with each of the peripheral tape nodes **1166** and communicates with the stationary gateway **1115** in accordance with a hierarchical wireless network of tape nodes. In some examples, each of the peripheral tape nodes **1166** includes a lower power wireless communications interface

and the master tape node **1160** is implemented by a tape node that includes a lower power communications interface for communicating with the peripheral tape nodes **1166** contained within the shipping container **1164**, and a higher power communications interface for communicating with the stationary gateway **1114**.

[0081] In some examples, when the doors of the shipping container **1164** are closed, the master tape node **1160** is operable to communicate wirelessly with the peripheral tape nodes **1166** contained within the shipping container **1164**. In an example, the master tape node **1160** is configured to collect sensor data from the peripheral tape nodes and, in some embodiments, process the collected data to generate, for example, one or more histograms from the collected data. When the doors of the shipping container **1164** are open, the master tape node **1160** is programmed to detect the door opening (e.g., with an accelerometer component of the master tape node **1160**) and, in addition to reporting the door opening event to the network service **1108**, the master tape node **1160** is further programmed to transmit the collected data and/or the processed data in one or more wireless messages to the stationary gateway **1114**. The stationary gateway **1114**, in turn, is operable to transmit the wireless messages received from the master tape node **1160** to the network service **1108** over the wireless network **1102**. Alternatively, in some examples, the stationary gateway **1114** also is operable to perform operations on the data received from the master tape node **1160** with the same type of data produced by the master node **1159** based on sensor data collected from the tape nodes **1142-1148**. In this way, the master tape node **1160** and the peripheral tape nodes **1166** create a hierarchical wireless network of nodes for transmitting, forwarding, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape nodes **1166** and the network service **1108** in a power-efficient and cost-effective way.

[0082] In some embodiments, the mobile gateways **1110** and/or the stationary gateways **1114** of the wireless tracking system **1100** are RFID/RF gateways **712, 714** as discussed in conjunction with FIG. 7 or are RFID/BLE gateways **812, 814** as discussed in conjunction with FIG. 8A-8B. As such, the mobile gateways **1110** and/or the stationary gateways **1114** are configured to communicate with RFID tags associated with respective assets of the wireless tracking system **1100** during various points of transport, storage, and the like.

[0083] In an example of the embodiment shown in FIG. 11, there are three classes of tape nodes: a short range tape node, a medium range tape node, and a long range tape node. The short range tape nodes typically are adhered directly to parcels containing assets. In the illustrated example, the tape nodes **1118, 1128, 1132, 1142-1148, 1166** are short range tape nodes. The short range tape nodes typically communicate with a low power wireless communication protocol (e.g., Bluetooth LE, Zigbee, or Z-wave). The medium range tape nodes typically are adhered to objects (e.g., a box **1126** and a shipping container **1160**) that are associated with multiple parcels or assets that are separated from the medium range tape nodes by a barrier or a large distance. In the illustrated example, the tape nodes **1124** and **1160** are medium range tape nodes. The medium range tape nodes typically communicate with a medium power wireless communication protocol (e.g., LoRa or Wi-Fi). The long-range tape nodes typically are adhered to mobile or stationary infrastructure of the wireless communication environment **1100**. In the illustrated example, the mobile gateway tape node **1112** and the stationary gateway tape node **1114** are long range tape nodes. The long range tape nodes typically communicate with other nodes using a high power wireless communication protocol (e.g., a cellular data communication protocol). In some examples, the mobile gateway tape node **1136** is adhered to a mobile vehicle (e.g., a truck). In these examples, the mobile gateway **1112** may be moved to different locations in the environment **1100** to assist in connecting other tape nodes to the server **1104**. In some examples, the stationary gateway tape node **1114** may be attached to a stationary structure (e.g., a wall) in the environment **1100** with a known geographic location. In these examples, other tape nodes in the environment can determine their geographic location by querying the gateway tape node **1114**.

Hierarchical Wireless Communications Network

[0084] FIG. 12 shows an example hierarchical wireless communications network of tape nodes **1270**. In this example, the short range tape node **1272** and the medium range tape node **1274** communicate with one another over their respective low power wireless communication interfaces **1276**, **1278**. The medium range tape node **1274** and the long range tape node **1280** communicate with one another over their respective medium power wireless communication interfaces **1278**, **1282**. The long range tape node **1280** and the network server **1204** communicate with one another over the high power wireless communication interface **1284**. In some examples, the low power communication interfaces **1276**, **1278** establish wireless communications with one another in accordance with the Bluetooth LE protocol, the medium power communication interfaces **1252**, **1282** establish wireless communications with one another in accordance with the LoRa communications protocol, and the high power communication interface **1284** establishes wireless communications with the server **1204** in accordance with a cellular communications protocol.

[0085] In some examples, the different types of tape nodes are deployed at different levels in the communications hierarchy according to their respective communications ranges, with the long range tape nodes generally at the top of the hierarchy, the medium range tape nodes generally in the middle of the hierarchy, and the short range tape nodes generally at the bottom of the hierarchy. In some examples, the different types of tape nodes are implemented with different feature sets that are associated with component costs and operational costs that vary according to their respective levels in the hierarchy. This allows system administrators flexibility to optimize the deployment of the tape nodes to achieve various objectives, including cost minimization, asset tracking, asset localization, and power conservation.

[0086] In some examples, a server **1204** of the network service **1208** designates a tape node at a higher level in a hierarchical communications network as a master node of a designated set of tape nodes at a lower level in the hierarchical communications network. For example, the designated master tape node may be adhered to a parcel (e.g., a box, pallet, or shipping container) that contains one or more tape nodes that are adhered to one or more packages containing respective assets. In order to conserve power, the tape nodes typically communicate according to a schedule promulgated by the server **1204** of the network service **1208**. The schedule usually dictates all aspects of the communication, including the times when particular tape nodes should communicate, the mode of communication, and the contents of the communication. In one example, the server **1204** transmits programmatic Global Scheduling Description Language (GSDL) code to the master tape node and each of the lower-level tape nodes in the designated set. In this example, execution of the GSDL code causes each of the tape nodes in the designated set to connect to the master tape node at a different respective time that is specified in the GSDL code, and to communicate a respective set of one or more data packets of one or more specified types of information over the respective connection. In some examples, the master tape node simply forwards the data packets to the server network node **1204**, either directly or indirectly through a gateway tape node (e.g., the long range tape node **1216** adhered to the mobile vehicle **1212** or the long range tape node **1214** adhered to an infrastructure component of the environment **1200**). In other examples, the master tape node processes the information contained in the received data packets and transmits the processed information to the server network node **1204**.

[0087] FIG. 13 shows an example method of creating a hierarchical communications network. In accordance with this method, a first tape node is adhered to a first asset in a set of associated assets, the first tape node including a first type of wireless communication interface and a second type of wireless communication interface having a longer range than the first type of wireless communication interface (FIG. 13, block **1390**). A second tape node is adhered to a second asset in the set, the second tape node including the first type of wireless communication interface, wherein the second tape node is operable to communicate with the first tape node over a wireless communication connection established between the first type of wireless communication interfaces of the first and second tape nodes (FIG. 13, block **1392**). An application executing on a computer

system (e.g., a server **1104** of a network service **1108**) establishes a wireless communication connection with the second type of wireless communication interface of the first tape node, and the application transmits programmatic code executable by the first tape node to function as a master tape node with respect to the second tape node (FIG. **13**, block **1394**).

[0088] In some embodiments, the second tape node is assigned the role of the master tape node with respect to the first tape node.

[0089] In some embodiments, a hierarchy is established with respect to wireless RFID/RF tags, RF tags, and other wireless tags of the wireless tracking system **1100**. For example, in embodiments comprising one or more RFID/RF tags and RF tags, a wireless tag or gateway tape node configured to broadcast an interrogation signal to the one or more RFID/RF tags or RF tags is a master tape node. In some embodiments, there may be one or more master tape nodes. The one or more master tape nodes may be assigned based on having a medium or high communications range as described above, or may be assigned based on other criteria, e.g., being affixed to the infrastructure of a vehicle or building so as to come into contact with or establish communications with a high number of assets. The one or more master tape nodes may receive information from the RFID/RF tags and RF tags responsive to the interrogation signal. In some embodiments, additional or different information may be received, e.g., sensor information describing an environment of a bin, container, or an asset within a container, and the like. The one or more master nodes may communicate the received information to a server or cloud of the wireless tracking system **1100** for processing, and may one or more processes and/or communicate subsets of the received information to the server or cloud of the wireless tracking system **1100** (e.g., identifying and transmitting information describing missing or compromised assets within a bin or container only).

Distributed Agent Operating System

[0090] As used herein, the term “node” refers to both a tape node and a non-tape node unless the node is explicitly designated as a “tape node” or a “non-tape node.” In some embodiments, a non-tape node may have the same or similar communication, sensing, processing and other functionalities and capabilities as the tape nodes described herein, except without being integrated into a tape platform. In some embodiments, non-tape nodes can interact seamlessly with tape nodes. Each node is assigned a respective unique identifier.

[0091] The following disclosure describes a distributed software operating system that is implemented by distributed hardware nodes executing intelligent agent software to perform various tasks or algorithms. In some embodiments, the operating system distributes functionalities (e.g., performing analytics on data or statistics collected or generated by nodes) geographically across multiple intelligent agents that are bound to items (e.g., parcels, containers, packages, boxes, pallets, a loading dock, a door, a light switch, a vehicle such as a delivery truck, a shipping facility, a port, a hub, etc.). In addition, the operating system dynamically allocates the hierarchical roles (e.g., master and slave roles) that nodes perform over time in order to improve system performance, such as optimizing battery life across nodes, improving responsiveness, and achieving overall objectives. In some embodiments, optimization is achieved using a simulation environment for optimizing key performance indicators (PKIs).

[0092] In some embodiments, the nodes are programmed to operate individually or collectively as autonomous intelligent agents. In some embodiments, nodes are configured to communicate and coordinate actions and respond to events. In some embodiments, a node is characterized by its identity, its mission, and the services that it can provide to other nodes. A node's identity is defined by its capabilities (e.g., battery life, sensing capabilities, and communications interfaces). A node's mission (or objective) is defined by the respective program code, instructions, or directives it receives from another node (e.g., a server or a master node) and the actions or tasks that it performs in accordance with that program code, instructions, or directives (e.g., sense temperature every hour and send temperature data to a master node to upload to a server). A node's services define the functions or tasks that it is permitted to perform for other nodes (e.g., retrieve temperature data

from a peripheral node and send the received temperature data to the server). At least for certain tasks, once programmed and configured with their identities, missions, and services, nodes can communicate with one another and request services from and provide services to one another independently of the server.

[0093] Thus, in accordance with the runtime operating system every agent knows its objectives (programmed). Every agent knows which capabilities/resources it needs to fulfill objective. Every agent communicates with every other node in proximity to see if it can offer the capability. Examples include communicate data to the server, authorize going to lower power level, temperature reading, send an alert to local hub, send location data, triangulate location, any boxes in same group that already completed group objectives.

[0094] Nodes can be associated with items. Examples of an item includes, but are not limited to for example, a package, a box, pallet, a container, a truck or other conveyance, infrastructure such as a door, a conveyor belt, a light switch, a road, or any other thing that can be tracked, monitored, sensed, etc. or that can transmit data concerning its state or environment. In some examples, a server or a master node may associate the unique node identifiers with the items.

[0095] Communication paths between tape and/or non-tape nodes may be represented by a graph of edges between the corresponding assets (e.g., a storage unit, truck, or hub). In some embodiments, each node in the graph has a unique identifier. A set of connected edges between nodes is represented by a sequence of the node identifiers that defines a communication path between a set of nodes.

[0096] Referring to FIG. 14A, a node **1420** (Node A) is associated with a package **1422** (Package A). In some embodiments, the node **1420** may be implemented as a tape node that is used to seal the package **1422** or it may be implemented as a label node that is used to label the package **1422**; alternatively, the node **1420** may be implemented as a non-tape node that is inserted within the package **1422** or embedded in or otherwise attached to the interior or exterior of the package **1422**. In the illustrated embodiment, the node **1420** includes a low power communications interface **1424** (e.g., a Bluetooth Low Energy communications interface). Another node **1426** (Node B), which is associated with another package **1430** (Package B), is similarly equipped with a compatible low power communications interface **1428** (e.g., a Bluetooth Low Energy communications interface).

[0097] In an example scenario, in accordance with the programmatic code stored in its memory, node **1426** (Node B) requires a connection to node **1420** (Node A) to perform a task that involves checking the battery life of Node A. Initially, Node B is unconnected to any other nodes. In accordance with the programmatic code stored in its memory, Node B periodically broadcasts advertising packets into the surrounding area. When the other node **1420** (Node A) is within range of Node B and is operating in a listening mode, Node A will extract the address of Node B and potentially other information (e.g., security information) from an advertising packet. If, according to its programmatic code, Node A determines that it is authorized to connect to Node B, Node A will attempt to pair with Node B. In this process, Node A and Node B determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path **1432** with Node A (e.g., a Bluetooth Low Energy formatted communication path), Node B determines Node A's identity information (e.g., master node), Node A's capabilities include reporting its current battery life, and Node A's services include transmitting its current battery life to other nodes. In response to a request from Node B, Node A transmits an indication of its current battery life to Node B.

[0098] Referring to FIG. 14B, a node **1434** (Node C) is associated with a package **1435** (Package C). In the illustrated embodiment, the Node C includes a low power communications interface **1436** (e.g., a Bluetooth Low Energy communications interface), and a sensor **1437** (e.g., a temperature sensor). Another node **1438** (Node D), which is associated with another package **1440** (Package D), is similarly equipped with a compatible low power communications interface **1442** (e.g., a Bluetooth Low Energy communications interface).

[0099] In an example scenario, in accordance with the programmatic code stored in its memory, Node D requires a connection to Node C to perform a task that involves checking the temperature in the vicinity of Node C. Initially, Node D is unconnected to any other nodes. In accordance with the programmatic code stored in its memory, Node D periodically broadcasts advertising packets in the surrounding area. When Node C is within range of Node D and is operating in a listening mode, Node C will extract the address of Node D and potentially other information (e.g., security information) from the advertising packet. If, according to its programmatic code, Node C determines that it is authorized to connect to Node D, Node C will attempt to pair with Node D. In this process, Node C and Node D determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path **1444** with Node C (e.g., a Bluetooth Low Energy formatted communication path), Node D determines Node C's identity information (e.g., a peripheral node), Node C's capabilities include retrieving temperature data, and Node C's services include transmitting temperature data to other nodes. In response to a request from Node D, Node C transmits its measured and/or locally processed temperature data to Node D.

[0100] Referring to FIG. **14C**, a pallet **1450** or other container or bin is associated with a master node **1451** that includes a low power communications interface **1452**, a GPS receiver **1454**, and a cellular communications interface **1456**. In some embodiments, the master node **1451** may be implemented as a tape node or a label node that is adhered to the pallet **1450**. In other embodiments, the master node **1451** may be implemented as a non-tape node that is inserted within the body of the pallet **1450** or embedded in or otherwise attached to the interior or exterior of the pallet **1450**. In other embodiments, the master node **1451** is a wireless tag with RFID/RF or BLE capabilities and is configured to transmit interrogation signals to one or more RFID/RF or RF tags associated with individual assets within the pallet **1450**.

[0101] The pallet **1450** provides a structure for grouping and containing packages **1459**, **1461**, **1463** each of which is associated with a respective peripheral node **1458**, **1460**, **1462** (Node E, Node F, and Node G). In the example of FIG. **14C**, packages **1459**, **1461**, **1463** are not necessarily related, and may have different destinations or be able to withstand different ambient conditions. In other examples, the pallet **1450** is an equivalent to a bin or container as discussed in conjunction with FIG. **1A-1B**, wherein assets within the pallet are related and identified using RFID or RF tags. Each of the peripheral nodes **1458**, **1460**, **1462** includes a respective low power communications interface **1464**, **1466**, **1468** (e.g., Bluetooth Low Energy communications interface). In the illustrated embodiment, each of the nodes E, F, G and the master node **1451** are connected to each of the other nodes over a respective low power communications path (shown by dashed lines).

[0102] In some embodiments as mentioned previously, the packages **1459**, **1461**, **1463** are grouped together because they are related. For example, the packages **1459**, **1461**, **1463** may share the same shipping itinerary or a portion thereof. In an example scenario, the master pallet node **1450** scans for advertising packets that are broadcasted from the peripheral nodes **1458**, **1460**, **1462**. In some examples, the peripheral nodes broadcast advertising packets during respective scheduled broadcast intervals. The master node **1451** can determine the presence of the packages **1459**, **1461**, **1463** in the vicinity of the pallet **1450** based on receipt of one or more advertising packets from each of the nodes E, F, and G. In some embodiments, in response to receipt of advertising packets broadcasted by the peripheral nodes **1458**, **1460**, **1462**, the master node **1451** transmits respective requests to the server to associate the master node **1451** and the respective peripheral nodes **1458**, **1460**, **1462**. In some examples, the master tape node requests authorization from the server to associate the master tape node and the peripheral tape nodes. If the corresponding packages **1459**, **1461**, **1463** are intended to be grouped together (e.g., they share the same itinerary or certain segments of the same itinerary), the server authorizes the master node **1451** to associate the peripheral nodes **1458**, **1460**, **1462** with one another as a grouped set of packages. In some embodiments, the server registers the master node and peripheral tape node identifiers with a group identifier. The server also may associate each node ID with a respective physical label ID that is affixed to the respective

package.

[0103] In some embodiments, after an initial set of packages is assigned to a multi-package group, the master node **1451** may identify another package arrives in the vicinity of the multi-package group. The master node may request authorization from the server to associate the other package with the existing multi-package group. If the server determines that the other package is intended to ship with the multi-package group, the server instructs the master node to merge one or more other packages with currently grouped set of packages. After all packages are grouped together, the server authorizes the multi-package group to ship. In some embodiments, this process may involve releasing the multi-package group from a containment area (e.g., customs holding area) in a shipment facility.

[0104] In some embodiments, the peripheral nodes **1458**, **1460**, **1462** include environmental sensors for obtaining information regarding environmental conditions in the vicinity of the associated packages **1459**, **1461**, **1463**. Examples of such environmental sensors include temperature sensors, humidity sensors, acceleration sensors, vibration sensors, shock sensors, pressure sensors, altitude sensors, light sensors, and orientation sensors.

[0105] In the illustrated embodiment, the master node **1451** can determine its own location based on geolocation data transmitted by a satellite-based radio navigation system **1470** (e.g., GPS, GLONASS, and NAVSTAR) and received by the GPS receiver **1454** component of the master node **1451**. In an alternative embodiment, the location of the master pallet node **1451** can be determined using cellular based navigation techniques that use mobile communication technologies (e.g., GSM, GPRS, CDMA, etc.) to implement one or more cell-based localization techniques. After the master node **1451** has ascertained its location, the distance of each of the packages **1459**, **1461**, **1463** from the master node **1451** can be estimated based on the average signal strength of the advertising packets that the master node **1451** receives from the respective peripheral node. The master node **1451** can then transmit its own location and the locations of the package nodes E, F, and G to a server over a cellular interface connection with a cell tower **1472**. Other methods of determining the distance of each of the packages **1459**, **1461**, **1463** from the master node **1451**, such as Received Signal-Strength Index (RSSI) based indoor localization techniques, also may be used.

[0106] In some embodiments, after determining its own location and the locations of the peripheral nodes, the master node **1451** reports the location data and the collected and optionally processed (e.g., either by the peripheral nodes peripheral nodes **1458**, **1460**, **1462** or the master node **1451**) sensor data to a server over a cellular communication path **1471** on a cellular network **1472**.

[0107] In some examples, nodes are able to autonomously detect logistics execution errors if packages that are supposed to travel together no longer travel together, and raise an alert. For example, a node (e.g., the master node **1451** or one of the peripheral nodes **1458**, **1460**, **1462**) alerts the server when the node determines that a particular package **1459** is being or has already been improperly separated from the group of packages. The node may determine that there has been an improper separation of the particular package **1459** in a variety of ways. For example, the associated node **1458** that is bound to the particular package **1459** may include an accelerometer that generates a signal in response to movement of the package from the pallet. In accordance with its intelligent agent program code, the associated node **1458** determines that the master node **1451** has not disassociated the particular package **1459** from the group and therefore broadcasts advertising packets to the master node, which causes the master node **1451** to monitor the average signal strength of the advertising packets and, if the master node **1451** determines that the signal strength is decreasing over time, the master node **1451** will issue an alert either locally (e.g., through a speaker component of the master node **1451**) or to the server.

[0108] Referring to FIG. **14D**, a truck **1480** is configured as a mobile node or mobile hub that includes a cellular communications interface **1482**, a medium power communications interface **1484**, and a low power communications interface **1486**. The communications interfaces **1480-586**

may be implemented on one or more tape and non-tape nodes. In an illustrative scenario, the truck **1480** visits a storage facility, such as a warehouse **1488**, to wirelessly obtain temperature data generated by temperature sensors in the medium range nodes **1490**, **1492**, **1494**. The warehouse **1488** contains nodes **1490**, **1492**, and **1494** that are associated with respective packages **1491**, **1493**, **1495**. In the illustrated embodiment, each node **1490-594** is a medium range node that includes a respective medium power communications interface **1496**, **2002**, **2008**, a respective low power communications interface **1498**, **2004**, **2010** and one or more respective sensors **2000**, **2006**, **2012**. In the illustrated embodiment, each of the package nodes **1490**, **1492**, **1494** and the truck **1480** is connected to each of the other ones of the package nodes through a respective medium power communications path (shown by dashed lines). In some embodiments, the medium power communications paths are LoRa formatted communication paths.

[0109] In some embodiments, the communications interfaces **1484** and **1486** (e.g., a LoRa communications interface and a Bluetooth Low Energy communications interface) on the node on the truck **1480** is programmed to broadcast advertisement packets to establish connections with other network nodes within range of the truck node. A warehouse **1488** includes medium range nodes **1490**, **1492**, **1494** that are associated with respective containers **1491**, **1493**, **1495** (e.g., packages, boxes, pallets, and the like). When the truck node's low power interface **1486** is within range of any of the medium range nodes **1490**, **1492**, **1494** and one or more of the medium range nodes is operating in a listening mode, the medium range node will extract the address of truck node and potentially other information (e.g., security information) from the advertising packet. If, according to its programmatic code, the truck node determines that it is authorized to connect to one of the medium range nodes **1490**, **1492**, **1494**, the truck node will attempt to pair with the medium range node. In this process, the truck node and the medium range node determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path with the truck node (e.g., a Bluetooth Low Energy formatted communication path **2014** or a LoRa formatted communication path **2017**), the truck node determines the identity information for the medium range node **1490** (e.g., a peripheral node), the medium range node's capabilities include retrieving temperature data, and the medium range node's services include transmitting temperature data to other nodes. Depending of the size of the warehouse **1488**, the truck **1480** initially may communicate with the nodes **1490**, **1492**, **1494** using a low power communications interface (e.g., Bluetooth Low Energy interface). If any of the anticipated nodes fails to respond to repeated broadcasts of advertising packets by the truck **1480**, the truck **1480** will try to communicate with the non-responsive nodes using a medium power communications interface (e.g., LoRa interface). In response to a request from the truck node **1484**, the medium range node **1490** transmits an indication of its measured temperature data to the truck node. The truck node repeats the process for each of the other medium range nodes **1492**, **1494** that generate temperature measurement data in the warehouse **1488**. The truck node reports the collected (and optionally processed, either by the medium range nodes **1490**, **1492**, **1494** or the truck node) temperature data to a server over a cellular communication path **2016** with a cellular network **2018**.

[0110] Referring to FIG. **14E**, a master node **2030** is associated with an item **2032** (e.g., a package) and grouped together with other items **2034**, **2036** (e.g., packages) that are associated with respective peripheral nodes **2038**, **2040**. The master node **2030** includes a GPS receiver **2042**, a medium power communications interface **2044**, one or more sensors **2046**, and a cellular communications interface **2048**. Each of the peripheral nodes **2038**, **2040** includes a respective medium power communications interface **2050**, **2052** and one or more respective sensors **2054**, **2056**. In the illustrated embodiment, the peripheral and master nodes are connected to one another over respective pairwise communications paths (shown by dashed lines). In some embodiments, the nodes **2030** **2038**, **2040** communicate through respective LoRa communications interfaces over LoRa formatted communications paths **2058**, **2060**, **2062**.

[0111] In the illustrated embodiment, the master and peripheral nodes **2038**, **2038**, **2040** include

environmental sensors for obtaining information regarding environmental conditions in the vicinity of the associated packages **2032**, **2034**, **2036**. Examples of such environmental sensors include temperature sensors, humidity sensors, acceleration sensors, vibration sensors, shock sensors, pressure sensors, altitude sensors, light sensors, and orientation sensors.

[0112] In accordance with the programmatic code stored in its memory, the master node **2030** periodically broadcasts advertising packets in the surrounding area. When the peripheral nodes **2038**, **2040** are within range of master node **2030**, and are operating in a listening mode, the peripheral nodes **2038**, **2040** will extract the address of master node **2030** and potentially other information (e.g., security information) from the advertising packets. If, according to their respective programmatic code, the peripheral nodes **2038**, **2040** determine that they are authorized to connect to the master node **2030**, the peripheral nodes **2038**, **2040** will attempt to pair with the master node **2030**. In this process, the peripheral nodes **2038**, **2040** and the master node and the peripheral nodes determine each other's identities, capabilities, and services. For example, after successfully establishing a respective communication path **2058**, **2060** with each of the peripheral nodes **2038**, **2040** (e.g., a LoRa formatted communication path), the master node **2030** determines certain information about the peripheral nodes **2038**, **2040**, such as their identity information (e.g., peripheral nodes), their capabilities (e.g., measuring temperature data), and their services include transmitting temperature data to other nodes.

[0113] After establishing LoRa formatted communications paths **2058**, **2060** with the peripheral nodes **2038**, **2040**, the master node **2030** transmits requests for the peripheral nodes **2038**, **2040** to transmit their measured and/or locally processed temperature data to the master node **2030**.

[0114] In the illustrated embodiment, the master node **2030** can determine its own location based on geolocation data transmitted by a satellite-based radio navigation system **2066** (e.g., GPS, GLONASS, and NAVSTAR) and received by the GPS receiver **2042** component of the master node **2030**. In an alternative embodiment, the location of the master node **2030** can be determined using cellular based navigation techniques that use mobile communication technologies (e.g., GSM, GPRS, CDMA, etc.) to implement one or more cell-based localization techniques. After the master node **2030** has ascertained its location, the distance of each of the packages **2034**, **2036** from the master node **2030** can be estimated based on the average signal strength of the advertising packets that the master node **2030** receives from the respective peripheral node. The master node **2030** can then transmit its own location and the locations of the package nodes E, F, and G to a server over a cellular interface connection with a cell tower **2072**. Other methods of determining the distance of each of the packages **2034**, **2036** from the master node **2030**, such as Received Signal-Strength Index (RSSI) based indoor localization techniques, also may be used.

[0115] In some embodiments, after determining its own location and the locations of the peripheral nodes, the master node **2030** reports the location data the collected and optionally processed (e.g., either by the peripheral nodes peripheral nodes **2034**, **2036** or the master node **2030**) sensor data to a server over a cellular communication path **2070** on a cellular network **2072**.

[0116] Referring to FIG. 15A, in some examples, each of one or more of the segments **1570**, **1572** of a tracking adhesive product **1574** includes a respective circuit **1575** that delivers power from the respective energy source **1576** to the respective tracking circuit **1578** (e.g., a processor and one or more wireless communications circuits) in response to an event. In some of these examples, the wake circuit **1575** is configured to transition from an off state to an on state when the voltage on the wake node **1577** exceeds a threshold level, at which point the wake circuit transitions to an on state to power-on the segment **1570**. In the illustrated example, this occurs when the user separates the segment from the tracking adhesive product **1574**, for example, by cutting across the tracking adhesive product **1574** at a designated location (e.g., along a designated cut-line **1580**). In particular, in its initial, un-cut state, a minimal amount of current flows through the resistors R.sub.1 and R.sub.2. As a result, the voltage on the wake node **1570** remains below the threshold turn-on level. After the user cuts across the tracking adhesive product **1574** along the designated

cut-line **1580**, the user creates an open circuit in the loop **1582**, which pulls the voltage of the wake node above the threshold level and turns on the wake circuit **1575**. As a result, the voltage across the energy source **1576** will appear across the tracking circuit **1578** and, thereby, turn on the segment **1570**. In particular embodiments, the resistance value of resistor R.sub.1 is greater than the resistance value of R.sub.2. In some examples, the resistance values of resistors R.sub.1 and R.sub.2 are selected based on the overall design of the adhesive product system (e.g., the target wake voltage level and a target leakage current).

[0117] In some examples, each of one or more of the segments of a tracking adhesive product includes a respective sensor and a respective wake circuit that delivers power from the respective energy source to the respective one or more of the respective tracking components **1578** in response to an output of the sensor. In some examples, the respective sensor is a strain sensor that produces a wake signal based on a change in strain in the respective segment. In some of these examples, the strain sensor is affixed to a tracking adhesive product and configured to detect the stretching of the tracking adhesive product segment as the segment is being peeled off a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a capacitive sensor that produces a wake signal based on a change in capacitance in the respective segment. In some of these examples, the capacitive sensor is affixed to a tracking adhesive product and configured to detect the separation of the tracking adhesive product segment from a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a flex sensor that produces a wake signal based on a change in curvature in the respective segment. In some of these examples, the flex sensor is affixed to a tracking adhesive product and configured to detect bending of the tracking adhesive product segment as the segment is being peeled off a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a near field communications sensor that produces a wake signal based on a change in inductance in the respective segment.

[0118] FIG. **15B** shows another example of a tracking adhesive product **1594** that delivers power from the respective energy source **1576** to the respective tracking circuit **1578** (e.g., a processor and one or more wireless communications circuits) in response to an event. This example is similar in structure and operation as the tracking adhesive product **1594** shown in FIG. **15A**, except that the wake circuit **1575** is replaced by a switch **1596** that is configured to transition from an open state to a closed state when the voltage on the switch node **1577** exceeds a threshold level. In the initial state of the tracking adhesive product **1594**, the voltage on the switch node is below the threshold level as a result of the low current level flowing through the resistors R.sub.1 and R.sub.2. After the user cuts across the tracking adhesive product **1594** along the designated cut-line **1580**, the user creates an open circuit in the loop **1582**, which pulls up the voltage on the switch node above the threshold level to close the switch **1596** and turn on the tracking circuit **1578**.

Exemplary Computer Apparatus

[0119] FIG. **16** shows an example embodiment of computer apparatus that is configured to implement one or more of the computing systems described in this specification. The computer apparatus **1620** includes a processing unit **1622**, a system memory **1624**, and a system bus **1626** that couples the processing unit **1622** to the various components of the computer apparatus **1620**. The processing unit **1622** may include one or more data processors, each of which may be in the form of any one of various commercially available computer processors. The system memory **1624** includes one or more computer-readable media that typically are associated with a software application addressing space that defines the addresses that are available to software applications. The system memory **1624** may include a read only memory (ROM) that stores a basic input/output system (BIOS) that contains start-up routines for the computer apparatus **1620**, and a random access memory (RAM). The system bus **1626** may be a memory bus, a peripheral bus or a local bus, and may be compatible with any of a variety of bus protocols, including PCI, VESA, Microchannel, ISA, and EISA. The computer apparatus **1620** also includes a persistent storage memory **1628** (e.g., a hard drive, a floppy drive, a CD ROM drive, magnetic tape drives, flash

memory devices, and digital video disks) that is connected to the system bus **1626** and contains one or more computer-readable media disks that provide non-volatile or persistent storage for data, data structures and computer-executable instructions.

[0120] A user may interact (e.g., input commands or data) with the computer apparatus **1620** using one or more input devices **1630** (e.g. one or more keyboards, computer mice, microphones, cameras, joysticks, physical motion sensors, and touch pads). Information may be presented through a graphical user interface (GUI) that is presented to the user on a display monitor **1632**, which is controlled by a display controller **1634**. The computer apparatus **1620** also may include other input/output hardware (e.g., peripheral output devices, such as speakers and a printer). The computer apparatus **1620** connects to other network nodes through a network adapter **1636** (also referred to as a “network interface card” or NIC).

[0121] A number of program modules may be stored in the system memory **1624**, including application programming interfaces **1638** (APIs), an operating system (OS) **1640** (e.g., the Windows® operating system available from Microsoft Corporation of Redmond, Washington U.S.A.), software applications **1641** including one or more software applications programming the computer apparatus **1620** to perform one or more of the steps, tasks, operations, or processes of the hierarchical classification systems described herein, drivers **1642** (e.g., a GUI driver), network transport protocols **1644**, and data **1646** (e.g., input data, output data, program data, a registry, and configuration settings).

[0122] Examples of the subject matter described herein, including the disclosed systems, methods, processes, functional operations, and logic flows, can be implemented in data processing apparatus (e.g., computer hardware and digital electronic circuitry) operable to perform functions by operating on input and generating output. Examples of the subject matter described herein also can be tangibly embodied in software or firmware, as one or more sets of computer instructions encoded on one or more tangible non-transitory carrier media (e.g., a machine readable storage device, substrate, or sequential access memory device) for execution by data processing apparatus.

[0123] The details of specific implementations described herein may be specific to particular embodiments of particular inventions and should not be construed as limitations on the scope of any claimed invention. For example, features that are described in connection with separate embodiments may also be incorporated into a single embodiment, and features that are described in connection with a single embodiment may also be implemented in multiple separate embodiments. In addition, the disclosure of steps, tasks, operations, or processes being performed in a particular order does not necessarily require that those steps, tasks, operations, or processes be performed in the particular order; instead, in some cases, one or more of the disclosed steps, tasks, operations, and processes may be performed in a different order or in accordance with a multi-tasking schedule or in parallel.

[0124] Other embodiments are within the scope of the claims.

Claims

1-7. (canceled)

8. A system comprising: one or more wireless tags, each of the one or more wireless tags associated with a respective asset of a plurality of assets; a wireless node associated with a container; and a gateway node configured to: broadcast an interrogation signal to the one or more wireless tags and the wireless node; receive one or more identifiers, each of the one or more identifiers corresponding to a respective wireless tag of the one or more wireless tags; receive a portion of a database of identifiers, the portion of the database comprising a plurality of identifiers associated with at least one of the received one or more identifiers; compare the received one or more identifiers to the plurality of identifier from the portion of the database; identify that the received one or more identifiers is missing at least one identifier of the plurality of associated identifiers;

determine an event associated with the plurality of assets; and transmit an alert of the determined event.

9. The wireless tracking system of claim 8, wherein the wireless node comprises a wireless communication device, the wireless communication device comprising: comprising a radio frequency identifier (RFID) circuit, a radio frequency (RF) circuit, a processor, a memory storing an RFID identifier (ID) associated with the RFID circuit, and an energy source electrically connected to the RFID circuit, the RF circuit, and the processor.

10. The system of claim 9, the memory further storing an RF identifier associated with the RF circuit.

11. The system of claim 8, at least one of the one or more wireless tags comprising an RFID tag.

12. The system of claim 8, wherein the interrogation signal comprises an RFID interrogation signal.

13. The system of claim 11, wherein the interrogation signal further comprises a wireless communication signal other than the RFID interrogation signal.

14. A system comprising: one or more wireless tags, each of the one or more wireless tags associated with a respective asset of a plurality of assets; a wireless node associated with a container; and a gateway node configured to: broadcast an interrogation signal to the one or more wireless tags and the wireless node; receive one or more tag identifiers from the one or more wireless tags; receive one or more container identifiers from the wireless node; associate the one or more received tag identifiers with the one or more received container identifier; and storing the associations between the one or more tag identifiers with the one or more container identifiers in a database.

15. A method comprising: broadcasting an interrogation signal to one or more wireless tags and a wireless node, each of the one or more wireless tags associated with an asset; responsive to the one or more tags receiving the interrogation signal, receiving one or more tag identifiers from the one or more wireless tags; responsive to the wireless node receiving the interrogation signal, receiving one or more node identifiers from the wireless node; associating the one or more received tag identifiers with the one or more received node identifiers; and storing the associations between the one or more tag identifiers with the one or more node identifiers in a database.
